

7 Some groups of living organisms produce neurotoxins.

Neurotoxins act at synapses, preventing the generation of nerve impulses.

Some scientists have been studying neurotoxins using a combination of computer science, statistics and mathematics.

(a) What is the name of a study that combines computer science, statistics and mathematics?

(1)

- ☐ **A** bioinformatics
- ☐ **B** computed tomography
- ☐ **C** epigenetics
- ☐ **D** forensic entomology

(b) Some neurotoxins act in these ways:

- inhibiting the release of acetylcholine
- blocking of ion channels
- blocking of acetylcholine receptors.

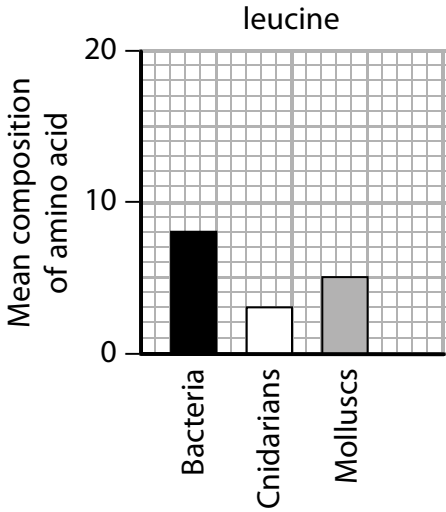
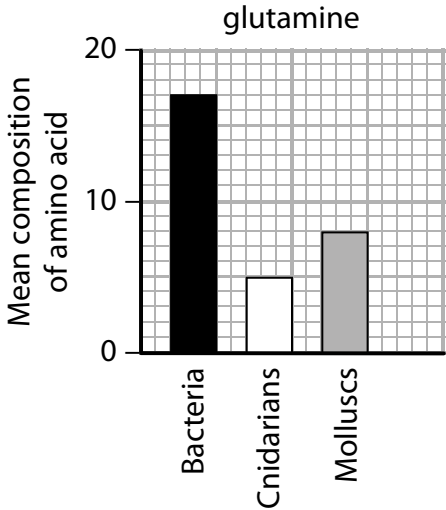
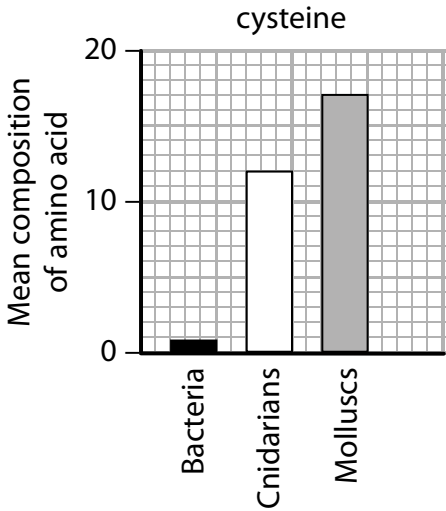
Explain how these neurotoxins prevent the generation of nerve impulses.

(3)

- (c) The table shows the mean composition of three amino acids, in neurotoxins and non-toxins.

Type of amino acid	Mean composition of amino acid in neurotoxins	Mean composition of amino acid in non-toxins
Cysteine	13.69	1.95
Glutamine	2.35	4.16
Leucine	5.53	10.17

The graphs show the mean composition of these amino acids in neurotoxins produced by three groups of organisms: bacteria, cnidarians and molluscs.



*(i) Comment on the amino acid composition of these neurotoxins. Use the information in the table and in the graphs to support your answer.

(6)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(ii) Explain why the replacement of one amino acid in a neurotoxin could decrease its toxicity.

(2)

.....

.....

.....

.....

.....

.....

(Total for Question 7 = 12 marks)
